Rongpeng He

PhD Candidate | Agricultural Remote Sensing · GeoAI · Earth Observation for Food Security

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RESEARCH VISION — AI-ENABLED EARTH INTELLIGENCE FOR FOOD SECURITY & SUSTAINABLE AGRIFOOD SYSTEMS

Develop scalable, generalizable, and trustworthy agricultural Earth intelligence by integrating multi-source Earth observation (EO), field observations, climate/soil/crop data, and geospatial AI. Current priorities include: (i) multimodal/self-supervised geospatial representation and foundation models; (ii) label-efficient and domain-generalizable crop/soil monitoring across Xinjiang–Central Asia; (iii) cloud-scale agricultural big-data pipelines; and (iv) operational indicators for crop area, cropping intensity, condition, yield, soil health, climate resilience, and SDG 2 decision support.

EDUCATION

2025–Present **University of Chinese Academy of Sciences / Aerospace Information Research Institute, CAS — PhD Candidate, Electronic Information (Remote Sensing)**

➤ Advisor: Prof. Jihua Meng. Doctoral work targets agricultural EO big data, cross-regional crop mapping, GeoAI, and food-security applications.

2022–2025 **University of Chinese Academy of Sciences / Aerospace Information Research Institute, CAS — M.Eng./M.Sc. in Electronic Information**

➤ GPA: 3.69/4.00; research in agricultural remote sensing, image processing, and soil nutrient monitoring.

2018–2022 **Chang'an University — B.Eng. in Surveying and Mapping Engineering**

➤ GPA: 3.65/5.00; ranked 4/98; recommended for postgraduate admission to UCAS.

SELECTED PUBLICATIONS

1. He, R.; Meng, J.; Na, H. **Dual-dimension spectral reconstruction: spatiotemporal mapping of soil organic matter under complex cropland surface conditions.** *International Journal of Digital Earth 19(2), 2697113 (2026).* DOI **FIRST AUTHOR**

2. He, R.; Meng, J.; Du, Y.; Lin, Z.; You, X.; Gao, X. **Enhancing Regional Topsoil Total Nitrogen Mapping Through Differentiated Fusion of Ground Hyperspectral Data and Satellite Images Under Low Vegetation Cover.** *Agriculture 14, 2145 (2024).* DOI **FIRST AUTHOR**

3. Meng, J.; He, R.; Lin, Z. **Status and Prospects of Google Earth Engine in Agricultural Management Research.** *Journal of Geo-information Science 26(4) (2024).* DOI **CO-AUTHOR**

DOCTORAL RESEARCH IN PROGRESS

➤ Data foundation: field crop-reference samples collected in Kashgar (southern Xinjiang, 2025) and northern Xinjiang (2026), combined with historical Xinjiang crop maps, Sentinel-1/2 time series, and climate/soil covariates.

➤ Methodological focus: historical weak supervision, multimodal phenological representation learning, self-supervised/geospatial-foundation-model embeddings, few-shot adaptation, domain generalization, and uncertainty-aware validation.

➤ Scaling objective: develop transferable agricultural mapping methods from southern-to-northern Xinjiang and further to data-scarce agricultural regions of Central Asia, especially Kyrgyzstan and Uzbekistan.

SELECTED RESEARCH & SUSTAINABLE-DEVELOPMENT PROJECTS

2025–2029 **Xinjiang Major Science & Technology Project — Satellite-UAV-Station Collaborative Monitoring of Wind-Sand and Salinity Hazards | Project Member (24 person-months)**

➤ Contribute to long-term multi-source EO data and parameter products for Xinjiang. The project targets 1.66 million km², 10–30 m products for 2000–2028, and 12+ environmental/agricultural variables including crop type, cropping intensity, soil and vegetation indicators.

2025–2027 **Kashgar Science & Technology Program — Aerospace Big-Data Resource/Environment Applications and Sustainable-Development Monitoring | Project Backbone**

➤ Lead/undertake cotton-parameter retrieval tasks and contribute to operational products for crop area, growth, cropping intensity, fallow land, and other agricultural indicators.

2023–2024 **National Key R&D-related Task — Integrated Software R&D for Unmanned-Farm Soil Parameter Retrieval | Major Participant**

➤ Developed multi-source remote-sensing workflows for six soil parameters. A differentiated ground-satellite fusion + Random Forest scheme achieved R^2=0.95 and RMSE=0.10 g/kg for topsoil total nitrogen mapping.

2024 **Big Earth Data in Support of the Sustainable Development Goals Report — SDG 2 "Zero Hunger" | Sustainable Agriculture Case Contributor**

➤ Contributed multi-source agricultural monitoring methods, including dynamic soil available-nitrogen monitoring and crop-maturity prediction; related outputs were incorporated into the report's sustainable-agriculture pathway case to support SDG 2 monitoring and decision support.

TECHNICAL & RESEARCH SKILLS

➤ Programming & geospatial data engineering: Python; NumPy, pandas, scikit-learn; GeoPandas, Rasterio, GDAL; reproducible batch workflows for large-area and long time-series EO data.

➤ Cloud & scalable EO: Google Earth Engine; multi-source data harmonization, feature engineering, cloud-scale batch processing, map production, and accuracy assessment.

➤ Remote sensing: Sentinel-1/2, Landsat, optical-SAR integration, hyperspectral processing, phenology/time-series analysis, spectral unmixing, and soil/crop parameter retrieval.

➤ AI / GeoAI: Random Forest; PyTorch; CNN/Transformer/Capsule-Network workflows; multimodal representation learning, transfer/domain adaptation, domain generalization, and few-shot learning.

➤ Research practice: field campaign and sample design; spatial cross-validation, error/uncertainty analysis; scientific visualization; academic writing and international oral presentation.

RESEARCH HIGHLIGHTS

➤ 2026 IJDE study: proposed a temporal-moisture-reduction + spatial-unmixing strategy and produced annual 10-m soil organic matter maps for 2015–2025; optimal RF performance reached mean $R^2 = 0.76 \pm 0.01$ and RMSE = 2.25 ± 0.03 g/kg.

➤ 2024 Agriculture study: developed differentiated fusion of enhanced multispectral image bands with Random Forest for regional topsoil total nitrogen mapping; $R^2 = 0.95$, RMSE = 0.10 g/kg, LCCC = 0.89.

➤ Operational orientation: experience linking field observations, satellite time series, cloud computing, and decision-oriented agricultural products for regional monitoring and sustainable-development reporting.

INTELLECTUAL PROPERTY & SOFTWARE

➤ Chinese Invention Patent ZL 2026 1 0297486.2, “Method, device and electronic equipment for monitoring soil organic matter in farmland,” granted May 19, 2026. Second inventor.

➤ Chinese Invention Patent ZL 2024 1 0904056.3, “Method and device for determining soil organic matter content through differential fine-registration of ground-satellite spectra,” granted Sep. 24, 2024. Second inventor.

➤ Farmland Soil Health Monitoring System V1.0, Software Copyright Registration No. 2024SR1254209, Aug. 2024.

SELECTED PRESENTATIONS

➤ Oral presentation, 23rd World Congress of Soil Science (WCSS 2026), Nanjing, China — regional soil total nitrogen mapping through differentiated ground-hyperspectral/satellite fusion.

➤ International Symposium on Earth Observation for Arid and Semi-Arid Environments (ISEO), Sep. 2024 — oral/session presentation on regional topsoil total nitrogen mapping.

FIELD, OPERATIONAL & SCIENCE-COMMUNICATION EXPERIENCE

➤ Southern Xinjiang crop survey (Kashgar, 2025): crop-reference sampling, field-survey design, geolocation/quality control, and linkage of parcel observations with satellite time series for classification and validation.

➤ Northern Xinjiang crop survey (2026): expanded reference collection across contrasting agroclimatic zones to support cross-regional transfer and generalization experiments.

➤ Kashgar operational monitoring: contributed to crop-area, condition, cropping-intensity, fallow-land, and cotton-parameter products for regional agricultural decision support.

➤ Science communication and training: participated in technical training on integrated satellite-UAV-ground agricultural monitoring for local technical teams and schools.

➤ End-to-end operational workflow: field observations → EO data engineering → cloud-scale modeling → map/product generation → accuracy assessment → decision-oriented reporting.

SELECTED AWARDS & HONORS

➤ Outstanding Presentation Award, 2025 Remote Sensing Science Graduate Forum.

➤ AIRCAS Enterprise Scholarship (2024–2025).

➤ UCAS First-Class Academic Scholarship (2023–2024); Second-Class Academic Scholarship (2022–2023).

➤ Meritorious Student, Outstanding Student Leader, and Outstanding League Secretary, UCAS (2022–2023).

➤ National First Prize, 2nd National College Student Surveying and Mapping Innovation Development Competition (Huace Navigation Cup), 2021.